Plan for Data Filter:

1. Which teammates will specify, design, and implement this microservice?

Nathan Zak and Mohana Vista will design this microservice; both they and Charles Gilbert will implement it.

2. Provide a **link** to the repository.

https://github.com/m-vista/DU-DataFilter/projects

3. Write at least three user stories for this Sprint.

First user story

(Front of index card) **Filter Data by String** *As a user, I want to be able to filter my data with a string so that I can quickly see the relevant entries.*
(Back of index card) Acceptance criteria Functional requirements • Given I'm filtering data, when I filter for a specific string then I will only see the entries that contain that exact string Quality attributes & Non-functional requirements • Performance: ◦ The service must filter every correlating entry from a dataset of 1000 entries in less than 2 seconds   ◦ Algorithm runs at O(n) or better

Second user story

(Front of index card)

Handle Invalid Filter Request

As a user, I want to be informed when no entries match my filter request so that I can adjust my request and try again, or work with that knowledge as a result of its own.

(Back of index card)

Acceptance criteria

Functional requirements

- Given I'm filtering my data, when I request a filter that doesn't match an entry in my dataset then I will receive an error message from the service that no results were found
- Given I'm filtering data, when I request an empty filter then I will receive an error message from the service telling me to enter criteria and try again

Quality attributes & Non-functional requirements

- Performance:
  - The service must filter every correlating entry from a dataset of 1000 entries in less than 2 seconds
  - Algorithm runs at O(n) or better

Third user story

(Front of index card)

Filter Data by Integer

As a user, I want to be able to filter my data by an integer or range of integers so that I can quickly see all the relevant entries.

(Back of index card)

Acceptance criteria

Functional requirements

- Given I'm filtering data, when I filter by a specific integer then I will see every entry with that exact integer
- Given I'm filtering data, when I filter by a specific range of integers then I will see every entry within the range of those integers

Quality attributes & Non-functional requirements

- Performance:
  - The service must filter every correlating entry from a dataset of 1000 entries in less than 2 seconds
  - Algorithm runs at O(n) or better

4. Take a **screenshot** that shows the user stories are in a Sprint Backlog in GitHub Projects.

   Screenshot

5. What kind of **communication pipe** will the microservice use? (e.g., text files, REST API)

   ZeroMQ sending JSON strings through REQ/REP sockets

6. How will other programs be able to **request data** from the microservice and what request parameters will be required?

JSON strings
Applications will request data through ZeroMQ REQ sockets using JSON string as parameters
create ZeroMQ request socket connect it to the filter service (port: xxxx) build a request message with: (key) is (value) (key) is (value) ... convert request message into JSON string send the JSON string through the socket, wait for a reply turn the reply from JSON back into data if the reply is "ok", use matching data else, show error message

7. How will other programs be able to **receive data** from the microservice and what data will the microservice provide?

JSON strings
A status field in a JSON string will either send “ok” and the correlating data or “error” and an explanatory message
if status is “ok”: read and display all relevant data else, show error message